# FUJITSU-MONAKA®

## Microarchitecture Manual

## Part of PMU Event

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# Preface

# 1. Performance Monitor Events

The processor is equipped with the performance monitoring unit (PMU) to monitor dynamic program behavior. In addition to the events defined in the ARMv9 Manual and the SVE Manual, the MONAKA has events unique to it. These events are designed to enable calculation of indicators that assist software performance analysis by not only using each event with its direct meaning but also combining multiple events. This chapter describes how to create these indicators. For details on the specifications of each event, see the MONAKA PMU Events. Some events count inaccurately. For details such event, see the MONAKA PMU Events Errata.

# 1.1. Instruction Mix

Table 1-1 summarizes events for calculating a dynamic instruction frequency distribution at the program execution time. All the events for Instruction Mix are designed to count architecture instruction commits. A group of instructions counted with these events has an inclusion relationship. The relationship means that the frequency of a group of instructions for which events are not defined can be calculated by combining those events. The inclusion relationship of events is represented by the levels of indentations in the event name column in Table 1-1. The rows that display "Other" do not refer to actual events but to items calculated by combining events. The formulas are shown in Table 1-2. Note that the SVE_MATH_SPEC event is not included in the INST_SPEC event.

**Table 1-1 Mix Target of Counting by Event**

| Event for Instruction | Mix Target of Counting by Event |
|---|---|
| INST_SPEC | All instructions |
| FP_SPEC | All instructions categorized as floating-point operation |
| FP_DIV_SPEC | All floating-point division instructions |
| ASE_FP_DIV_SPEC | Advanced SIMD floating-point division Instruction |
| SVE_FP_DIV_SPEC | SVE Floating-Point division Instructions |
| other (Basic FP DIV operations) | General floating-point division instructions |
| FP_SQRT_SPEC | All floating-point square root operations |
| ASE_FP_SQRT_SPEC | Advanced SIMD floating-point Square Root Operation Instruction |
| SVE_FP_SQRT_SPEC | SVE floating point square root operation instructions |
| other (Basic FP SQRT operations) | General floating point square root instructions |
| FP_FMA_SPEC | All FMA operation instructions |
| ASE_FP_FMA_SPEC | Advanced SIMD FMA-based operation instructions |
| SVE_FP_FMA_SPEC | SVE FMA-based operation instruction |
| other (Basic FP FMA operations) | General FMA operation instructions |
| FP_MUL_SPEC | All floating-point multiply instructions |
| ASE_FP_MUL_SPEC | Advanced SIMD Floating-Point multiply Instruction |
| SVE_FP_MUL_SPEC | SVE floating-Point multiply Instruction |
| other (Basic FP MUL operations) | General floating-point multiply instructions |
| FP_ADDSUB_SPEC | All floating-point addition and subtraction instructions |
| ASE_FP_ADDSUB_SPEC | Advanced SIMD floating-point addition and subtraction instructions |
| SVE_FP_ADDSUB_SPEC | SVE floating-point addition and subtraction instructions |
| other (Basic FP ADDSUB operations) | General floating-point addition and subtraction instructions |
| FP_RECPE_SPEC | All floating-point and reciprocal square root approximation instructions |
| ASE_FP_RECPE_SPEC | Advanced SIMD floating-point and reciprocal square root approximation instructions |
| SVE_FP_RECPE_SPEC | SVE floating-point and reciprocal square root approximation instructions |
| ASE_SVE_FP_VREDUCE_SPEC | All floating-point reduction operation instructions (except Accumulating operations) |
| ASE_FP_VREDUCE_SPEC | Advanced SIMD floating-point reduction operation instructions |
| SVE_FP_VREDUCE_SPEC | SVE floating-point reduction operation instructions |
| SVE_FP_AREDUCE_SPEC | SVE accumulating floating-point reduction operation instructions |
| SVE_FP_PREDUCE_SPEC | SVE floating-point pairwise add operation instructions |

| Event for Instruction | Mix Target of Counting by Event |
|---|---|
| ASE_SVE_FP_DOT_SPEC | Advanced SIMD and SVE floating-point Dot-product instructions |
| ASE_SVE_FP_MMLA_SPEC | Advanced SIMD and SVE floating-point matrix-multiply instructions |
| other (Basic FP operations) | General floating-point instructions |
| FP_CVT_SPEC | Floating-point precision conversion instructions (including conversion to values in generic registers) |
| ASE_FP_CVT_SPEC | Advanced SIMD floating-point precision conversion instructions |
| SVE_FP_CVT_SPEC | SVE floating-point precision conversion instructions |
| other (Basic FP CVT operations) | General floating-point precision conversion instructions |
| FP_MV_SPEC | Transfer instructions using floating-point registers (including general-purpose registers) |
| INT_SPEC | All instructions classified as fixed-point instructions |
| INT_DIV_SPEC | All fixed-point division instruction |
| SVE_INT_DIV_SPEC | SVE fixed-point division instruction |
| SVE_INT_DIV64_SPEC | SVE 64 bits fixed-point division instruction |
| other (Basic SVE INT DIV operations) | SVE general fixed-point division instruction |
| other(Basic V9 INT DIV operations) | General fixed-point division instruction |
| INT_MUL_SPEC | All fixed-point multiply instructions |
| ASE_INT_MUL_SPEC | Advanced SIMD fixed-point multiply instructions |
| SVE_INT_MUL_SPEC | SVE fixed-point multiply instructions |
| SVE_INT_MUL64_SPEC | SVE 64 bits fixed-point multiply instructions |
| SVE_INT_MULH64_SPEC | SVE 64 bits fixed-point multiply instructions ([SU]MULH) |
| other(Basic SVE INT MUL64, not narrowing operations) | General SVE 64 bits fixed-point multiply instructions |
| other (Basic V9 INT MUL operations) | General fixed-point multiply instructions |
| ASE_SVE_INT_VREDUCE_SPEC | All fixed-point reduction instructions |
| ASE_INT_VREDUCE_SPEC | Advanced SIMD fixed-point reduction instructions |
| SVE_INT_VREDUCE_SPEC | SVE fixed-point reduction instructions |
| ASE_SVE_INT_DOT_SPEC | Advanced SIMD and SVE fixed-point Dot-product instructions |
| ASE_SVE_INT_MMLA_SPEC | Advanced SIMD and SVE fixed-point matrix-multiply instructions |
| other (Basic INT operations) | General fixed-point reduction instructions |
| PRD_SPEC | Instructions using Predicate register |
| LD_SPEC | All load instructions |
| BASE_LD_REG_SPEC | All load instructions to general-purpose register |
| LDREX_SPEC | load-exclusive instructions |
| other (Basic int register loads) | General load instructions to general-purpose register |
| ASE_SVE_LD_SPEC | All load instructions to floating-point register (Load instructions of Advanced SIMD and SVE) |
| FP_LD_SPEC | Scalar load instructions to floating-point register |
| other (All vector loads) | Vector load instructions to floating-point register |
| SVE_LDR_REG_SPEC | All SVE LDR instructions |
| SVE_LDR_PREG_SPEC | SVE LDR (predicate) instructions |
| other (LDR vector) | SVE LDR (vector) instructions |
| BC_LD_SPEC | Advanced SIMD and SVE Replicate and broadcast load instructions to floating-point register (LD1R) |
| ASE_SVE_LD_MULTI_SPEC | Advanced SIMD and SVE Multiple structure load instructions to floating-point register (LD[234]*) |
| SVE_LD_GATHER_SPEC | SVE Gather load instructions |
| SVE_LDFF_SPEC | SVE first-fault and non-fault load instructions |
| SVE_LDNT_CONTIG_SPEC | SVE non-temporal load instructions |

| Event for Instruction | Mix Target of Counting by Event |
|---|---|
| other (Basic vector loads) | General vector load instructions to floating-point register |
| ST_SPEC | All store instructions |
| BASE_ST_REG_SPEC | All store instructions from general-purpose register |
| STREX_SPEC | store-exclusive instructions |
| other (Basic int register stores) | General store instructions from general-purpose register |
| ASE_SVE_ST_SPEC | All store instructions from floating-point register (Store instructions of Advanced SIMD and SVE) |
| FP_ST_SPEC | Scalar store instructions from floating-point register |
| other (All vector stores) | Vector store instructions from floating-point register |
| SVE_STR_REG_SPEC | All SVE STR instructions |
| SVE_STR_PREG_SPEC | SVE STR (predicate) instructions |
| other (STR vector) | SVE STR (vector) instructions |
| ASE_SVE_ST_MULTI_SPEC | Advanced SIMD and SVE Multiple structure store from floating-point register (ST[234]*) |
| SVE_ST_SCATTER_SPEC | SVE Scatter store instructions |
| SVE_STNT_CONTIG_SPEC | SVE non-temporal store instructions |
| other (Basic vector stores) | General vector store instructions from floating-point register |
| PRF_SPEC | All prefetch instructions |
| SVE_PRF_GATHER_SPEC | SVE gather prefetch instructions |
| SVE_PRF_CONTIG_SPEC | SVE contiguous prefetch instructions |
| other (Prefetch in base instruction) | Base instruction prefetch |
| DCZVA_SPEC | DC ZVA instructions |
| BR_PRED | All branch instructions |
| BR_IMMED_SPEC | Conditional and unconditional branch instructions (immediate) |
| BR_RETURN_SPEC | Return instructions |
| BR_INDIRECT_SPEC | Indirect branch instructions (excluding return instructions) |
| CRYPTO_SPEC | All encryption instructions |
| SVE_MOVPRFX_SPEC | All SVE MOVPRFX instructions |
| SVE_MOVPRFX_Z_SPEC | SVE MOVPRFX (Zeroing) instructions |
| SVE_MOVPRFX_M_SPEC | SVE MOVPRFX (Merging) instructions |
| SVE_MOVPRFX_U_SPEC | SVE MOVPRFX instructions (not fused with the prefixed instruction) |
| ISB_SPEC | ISB instructions |
| DSB_SPEC | DSB instructions |
| DMB_SPEC | DMB instructions |
| CSDB_SPEC | CSDB instructions |
| CID_WRITE_RETIRED | Writing to CONTEXTIDR_EL1 register |
| other (Base insts. excluding load/store) | Instructions that belong to Base Instructions excluding load/store instructions |
| **Event Independent of Inclusion Relationship** | |
| SVE_MATH_SPEC | SVE mathematical function auxiliary instructions |

**Table 1-2 Event Independent of Inclusion Relationship**

| Item | Formula |
|---|---|
| Basic FP DIV operations | FP_DIV_SPEC – (ASE_FP_DIV_SPEC + SVE_FP_DIV_SPEC) |
| Basic FP SQRT operations | FP_SQRT_SPEC – (ASE_FP_SQRT_SPEC + SVE_FP_SQRT_SPEC) |
| Basic FP FMA operations | FP_FMA_SPEC – (ASE_FP_FMA_SPEC + SVE_FP_FMA_SPEC) |
| Basic FP MUL operations | FP_MUL_SPEC – (ASE_FP_MUL_SPEC + SVE_FP_MUL_SPEC) |
| Basic FP ADDSUB operations | FP_ADDSUB_SPEC – (ASE_FP_ADDSUB_SPEC + SVE_FP_ADDSUB_SPEC) |

| Item | Formula |
|---|---|
| Basic FP operations | FP_SPEC<br>– (FP_DIV_SPEC + FP_SQRT_SPEC + FP_FMA_SPEC + FP_MUL_SPEC<br>+ FP_ADDSUB_SPEC + FP_RECPE_SPEC + ASE_SVE_FP_VREDUCE_SPEC<br>+ SVE_FP_AREDUCE_SPEC + SVE_FP_PREDUCE_SPEC + ASE_SVE_FP_DOT_SPEC + ASE_SVE_FP_MMLA_SPEC) |
| Basic FP CVT operations | FP_CVT_SPEC – (ASE_FP_CVT_SPEC + SVE_FP_CVT_SPEC) |
| Basic SVE INT DIV operations | SVE_INT_DIV_SPEC - SVE_INT_DIV64_SPEC |
| Basic V9 INT DIV operations | INT_DIV_SPEC - SVE_INT_DIV_SPEC |
| Basic SVE INT MUL64,<br>not narrowing operations | SVE_INT_MUL64_SPEC - SVE_INT_MULH64_SPEC |
| Basic V9 INT MUL operations | INT_MUL_SPEC – (ASE_INT_MUL_SPEC + SVE_INT_MUL_SPEC) |
| Basic INT operations | INT_SPEC<br>– (INT_DIV_SPEC + INT_MUL_SPEC + ASE_SVE_INT_VREDUCE_SPEC<br>+ ASE_SVE_INT_DOT_SPEC + ASE_SVE_INT_MMLA_SPEC) |
| Basic int register loads | BASE_LD_REG_SPEC - LDREX_SPEC |
| All vector loads | ASE_SVE_LD_SPEC – FP_LD_SPEC |
| LDR vector | SVE_LDR_REG_SPEC – SVE_LDR_PREG_SPEC |
| Basic vector loads | ASE_SVE_LD_SPEC<br>– (FP_LD_SPEC + SVE_LDR_REG_SEPC + BC_LD_SPEC + ASE_SVE_LD_MULTI_SPEC<br>+ SVE_LD_GATHER_SPEC + SVE_LDFF_SPEC + SVE_LDNT_CONTIG_SPEC) |
| Basic int register stores | BASE_ST_REG_SPEC - STREX_SPEC |
| All vector stores | ASE_SVE_ST_SPEC – FP_ST_SPEC |
| STR vector | SVE_STR_REG_SPEC – SVE_STR_PREG_SPEC |
| Basic vector stores | ASE_SVE_ST_SPEC<br>– (FP_ST_SPEC + SVE_STR_REG_SPEC + ASE_SVE_ST_MULTI_SPEC<br>SVE_ST_SCATTER_SPEC + SVE_STNT_CONTIG_SPEC) |
| Prefetch in base instruction | PRF_SPEC<br>– (SVE_PRF_GATHER_SPEC + SVE_PRF_CONTIG_SPEC) |
| Base insts. excluding load/store | INST_SPEC<br>– (FP_SPEC + FP_CVT_SPEC + FP_MV_PSEC + INT_SPEC + PRD_SPEC<br>+ LD_SPEC + ST_SPEC + PRF_SPEC + DCZVA_SPEC + BR_PRED<br>+ CRYPTO_SPEC + SVE_MOVPRFX_SPEC + ISB_SPEC + DSB_SPEC + DMB_SPEC<br>+ CSDB_SPEC + CID_WRITE_RETIRED) |

# 1.2. FLOPS

Table 1-3 summarizes events for calculating floating operations per second (FLOPS). These events count the number of floating-point operations that are committed. Among these events, the ones related to SVE instructions only count the number of operations in units of 128 bits, so they are not affected by vector length at the program execution. Therefore, the correct number of operations must be calculated with consideration of vector length at the execution. For example, if the Vector Length is measured at 256 bits, the measured value is halved and the measurer must double it. In addition, operation of FMA instruction are counted as two operations per element.

Since the PMU is a resource in units of PEs, the number of operations measured by the events is in units of PEs. Therefore, calculations of the total number of operations at the parallel execution must consider the job model, etc. Since FLOPS is the number of operations per unit time, external parameters (processor operation frequency during program execution and program execution time) are separately required. To calculate a highly precise execution time, we recommend using the CPU_CYCLES event obtained at the same time.

**Table 1-3 Performance Events for FLOPS**

| Performance Event | Description of Event |
|---|---|
| FP_SCALE_OPS_SPEC | Number of operations per 128 bits, taking into account the number of elements in each instruction of the SVE instruction |
| FP_FIXED_OPS_SPEC | Number of operations taking into account the number of element in SIMD&FP instruction |
| FP_FP8_SCALE_MIN_OPS_SPEC | Number of FP_SCALE_OPS_SPEC operations, smallest type is 8-bit floating-point. |
| FP_FP8_FIXED_MIN_OPS_SPEC | Number of FP_FIXED_OPS_SPEC operations, smallest type is 8-bit floating-point. |
| FP_BF16_SCALE_MIN_OPS_SPEC | Number of FP_SCALE_OPS_SPEC operations, smallest type is BFloat16 floating-point. |
| FP_BF16_FIXED_MIN_OPS_SPEC | Number of FP_FIXED_OPS_SPEC operations, smallest type is BFloat16 floating-point. |
| FP_HP_SCALE_OPS_SPEC | Number of FP_SCALE_OPS_SPEC operations with only semi-precision operations |
| FP_HP_FIXED_OPS_SPEC | Number of FP_FIXED_OPS_SPEC operations with only semi-precision operations |
| FP_SP_SCALE_OPS_SPEC | Number of FP_SCALE_OPS_SPEC operations with only single-precision operations |
| FP_SP_FIXED_OPS_SPEC | Number of FP_FIXED_OPS_SPEC operations with only single-precision operations |
| FP_DP_SCALE_OPS_SPEC | Number of FP_SCALE_OPS_SPEC operations for double-precision operations only |
| FP_DP_FIXED_OPS_SPEC | Number of FP_FIXED_OPS_SPEC operations that are double-precision operations only |
| INT_SCALE_OPS_SPEC | Number of fixed-point operations counted by SVE_INT_SPEC |
| INT_FIXED_OPS_SPEC | Number of fixed-point operations not counted by SVE_INT_SPEC |

# 1.3. Hardware Resource Monitor Events

Table 1-4 summarizes events for monitoring the behavior of basic processor resources. These events count dynamic hardware behaviors, such as cache misses and branch misprediction , at the program execution.

**Table 1-4 Performance Events for Hardware Resource Monitoring**

| Performance Event | Description of Event |
|---|---|
| BR_MIS_PRED | Number of times of pipeline flush caused by a branch misprediction |
| L1I_CACHE_REFILL | Number of L1I cache misses |
| L1I_CACHE_REFILL_DM_RD | Number of L1I cache misses due to demand read access |
| L1I_CACHE_REFILL_PRF | Number of L1I cache misses due to hardware/software prefetch access |
| L1I_CACHE_REFILL_HWPRF | Number of L1I cache misses due to hardware prefetch access |
| L1I_CACHE_HIT | Number of L1I cache hit |
| L1I_CACHE_HIT_RD | Number of L1I cache hit read access |
| L1I_LFB_HIT_RD | Number of cache-line hit due to read access during L1I cache fill |
| L1D_CACHE_REFILL | Number of L1D cache misses |
| L1D_CACHE_BTC | Number of L1D cache access due to demand access changing L1D cache status (shared → exclusive) |
| L1D_CACHE_REFILL_DM | Number of L1D cache misses due to demand access |
| L1D_CACHE_REFILL_DM_RD | Number of L1D cache misses due to demand read access |
| L1D_CACHE_REFILL_DM_WR | Number of L1D cache misses due to demand write access |
| L1D_CACHE_REFILL_RD | Number of L1D cache misses due to read access |
| L1D_CACHE_REFILL_WR | Number of L1D cache misses due to write access |
| L1D_CACHE_REFILL_PRF | Number of L1D cache misses due to hardware/software prefetch access |
| L1D_CACHE_REFILL_HWPRF | Number of L1D cache misses due to hardware prefetch access |
| L1D_CACHE_WB | Number of times of writeback from the L1D cache |
| L1D_CACHE_HIT | Number of L1D cache hit |
| L1D_CACHE_HIT_RD | Number of L1D cache hit due to demand read access |
| L1D_CACHE_HIT_WR | Number of L1D cache hit due to demand write access |
| L1D_LFB_HIT_RD | Number of cache-line hit due to demand read access during L1D fill |
| L1D_LFB_HIT_WR | Number of cache-line hit due to demand write access during L1D fill |
| L2D_CACHE_REFILL | Number of L2 cache misses |
| L2D_CACHE_BTC | Number of L2 cache access due to demand access changing L1D and L2 cache status (shared → exclusive) |
| L2D_CACHE_REFILL_DM | Number of L2 cache misses due to demand flow |
| L2D_CACHE_REFILL_DM_RD | Number of L2 cache misses due to demand read flow |
| L2D_CACHE_REFILL_DM_WR | Number of L2 cache misses due to demand write flow |
| L2D_CACHE_REFILL_DM_WR_EXCL | Number of L2 cache misses due to demand write flow (exclusive monitor) |
| L2D_CACHE_REFILL_DM_WR_ATOM | Number of L2 cache misses due to demand write flow (atomic operation) |
| L2D_CACHE_REFILL_RD | Number of L2 cache misses due to read flow |
| L2D_CACHE_REFILL_WR | Number of L2 cache misses due to write flow |
| L2D_CACHE_REFILL_PRF | Number of L2 cache misses due to prefetch flow |
| L2D_CACHE_REFILL_HWPRF | Number of L2 cache misses due to prefetch flow which was generated by hardware prefetch |

| Performance Event | Description of Event |
|---|---|
| L2D_CACHE_WB | Number of times of writeback from the L2 cache |
| L2D_CACHE_HIT | Number of L2 cache hit |
| L2D_CACHE_HIT_RD | Number of L2 cache hit due to demand read flow |
| L2D_CACHE_HIT_WR | Number of L2 cache hit due to demand write flow |
| L2D_LFB_HIT_RD | Number of L2 cache hit due to demand read flow during L2 fill |
| L2D_LFB_HIT_WR | Number of L2 cache hit due to demand write flow during L2 fill |
| L2D_CACHE_REFILL_L3D_MISS | Number of L3 cache misses |
| L2D_CACHE_REFILL_L3D_MISS_DM | Number of L3 cache misses due to demand flow |
| L2D_CACHE_REFILL_L3D_MISS_DM_RD | Number of L3 cache misses due to demand read flow |
| L2D_CACHE_REFILL_L3D_MISS_DM_WR | Number of L3 cache misses due to demand write flow |
| L2D_CACHE_REFILL_L3D_MISS_PRF | Number of L3 cache misses due to prefetch flow |
| L2D_CACHE_REFILL_L3D_MISS_HWPRF | Number of L3 cache misses due to prefetch flow generated by hardware prefetch |
| L2D_CACHE_REFILL_L3D_HIT | Number of L3 cache hit |
| L2D_CACHE_REFILL_L3D_HIT_DM | Number of L3 cache hit due to demand flow |
| L2D_CACHE_REFILL_L3D_HIT_DM_RD | Number of L3 cache hit due to demand read flow |
| L2D_CACHE_REFILL_L3D_HIT_DM_WR | Number of L3 cache hit due to demand write flow |
| L2D_CACHE_REFILL_L3D_HIT_PRF | Number of L3 cache hit due to prefetch flow |
| L2D_CACHE_REFILL_L3D_HIT_HWPRF | Number of L3 cache hit due to prefetch flow which was generated by hardware prefetch |
| L2D_CACHE_REFILL_L3D_MISS_DM_PFTGT_HIT | Number of L3 cache misses and prefetch buffer hit due to demand flow |
| L2D_CACHE_REFILL_L3D_MISS_DM_RD_PFTGT_HIT | Number of L3 cache misses and prefetch buffer hit due to demand read flow |
| L2D_CACHE_REFILL_L3D_MISS_DM_WR_PFTGT_HIT | Number of L3 cache misses and prefetch buffer hit due to demand write flow |
| L2D_CACHE_REFILL_L3D_MISS_DM_L_MEM | Number of L3 cache misses due to demand flow accessing memory in the same socket |
| L2D_CACHE_REFILL_L3D_MISS_DM_FR_MEM | Number of L3 cache misses due to demand flow accessing memory in different socket |
| L2D_CACHE_REFILL_L3D_MISS_DM_L_L2 | Number of L3 cache misses due to demand flow where the requests access the different L2 cache from the request in the same Cluster as the requests in the same COD/SRD set as the requests |
| L2D_CACHE_REFILL_L3D_MISS_DM_NR_L2 | Number of L3 cache misses due to demand flow where the requests access the L2 cache in the different COD/SRD set from the request in the same socket as the requests or in the different Cluster from the request in the same COD/SRD set as the requests |
| L2D_CACHE_REFILL_L3D_MISS_DM_NR_L3 | Number of L3 cache misses due to demand flow where the requests access the L3 cache in the different COD/SRD set from the request in the same socket as the requests or in the different Cluster from the request in the same COD/SRD set as the requests |
| L2D_CACHE_REFILL_L3D_MISS_DM_FR_L2 | Number of L3 cache misses due to demand flow accessing L2 cache in different sockets |
| L2D_CACHE_REFILL_L3D_MISS_DM_FR_L3 | Number of L3 cache misses due to demand flow accessing L3 cache in different sockets |
| L1I_CACHE_REFILL_PERCYC | Cumulative number of cycles for L1I cache misses |
| L1D_CACHE_REFILL_PERCYC | Cumulative number of cycles for L1D cache misses |
| L2D_CACHE_REFILL_PERCYC | Cumulative number of cycles for L2 cache misses |
| L1I_TLB_REFILL | Number of L1-ITLB misses |
| L1D_TLB_REFILL | Number of L1-DTLB misses |
| L2I_TLB_REFILL | Number of L2-ITLB misses |
| L2D_TLB_REFILL | Number of L2-DTLB misses |

| Performance Event | Description of Event |
|---|---|
| EFFECTIVE_INST_SPEC | Number of architecture instructions committed, excluding MOVPRFX instructions |
| EAGA_VAL | Number of cycles used by EAGA pipeline |
| EAGB_VAL | Number of cycles used by EAGB pipeline |
| PRX_VAL | Number of cycles used by PRX pipeline |
| EXA_VAL | Number of cycles used by the EXA pipeline |
| EXB_VAL | Number of cycles used by the EXB pipeline |
| EXC_VAL | Number of cycles used by the EXC pipeline |
| EXD_VAL | Number of cycles used by the EXD pipeline |
| FLA_VAL | Number of cycles used by the FLA pipeline |
| FLB_VAL | Number of cycles used by the FLB pipeline |
| STEA_VAL | Number of cycles used by the STEA pipeline |
| STEB_VAL | Number of cycles used by the STEB pipeline |
| STFL_VAL | Number of cycles used by the STFL pipeline |
| STPX_VAL | Number of cycles used by the STPX pipeline |
| BR_PRED | Number of branch instructions committed |
| CPU_CYCLES | Number of cycles in the core |
| L1_PIPE_COMP_GATHER_2FLOW | Number of times where 2 elements of the gather instructions became 2-flows because 2 elements could not be combined. |
| L1_PIPE_COMP_GATHER_1FLOW | Number of times where 2 elements of the gather instructions became 1-flow because 2 elements could be combined. |
| L1_PIPE_COMP_GATHER_0FLOW | Number of times where 2 elements of the gather instructions became 0-flow because both predicate values are 0. |
| L1_PIPE_COMP_SCATTER_1FLOW | Number of flows of the scatter instructions. |
| L1_PIPE0_VAL | Number of valid cycles of L1D cache pipeline#0. |
| L1_PIPE1_VAL | Number of valid cycles of L1D cache pipeline#1. |
| L1_PIPE2_VAL | Number of valid cycles of L1D cache pipeline#2. |
| L2_PIPE_VAL | Number of valid cycles of L2 cache pipeline. |
| L1_PIPE0_VAL_IU_NOT_SEC0 | Number of requests in L1D cache pipeline#0 that its sector cache ID is not 0. |
| L1_PIPE1_VAL_IU_NOT_SEC0 | Number of requests in L1D cache pipeline#1 that its sector cache ID is not 0. |
| OP_RETIRED | Number of every architecturally executed micro-operation. |
| OP_SPEC | Number of every speculatively executed micro-operation. |

By using the Event inTable 1-4, the performance index of the hardware at the time of program execution can be calculated. Summarize in Table 1-5.

**Table 1-5 Method to Calculate Hardware Performance Indicators at Program Execution**

| Indicator | Formula |
|---|---|
| Cycles per Instruction (CPI) | CPU_CYCLES / EFFECTIVE_INST_SPEC |
| Branch misprediction rate | BR_MIS_PRED / EFFECTIVE_INST_SPEC |
| L1I cache miss rate | L1I_CACHE_REFILL / EFFECTIVE_INST_SPEC |
| L1I cache miss rate due to demand read access | L1I_CACHE_REFILL_DM_RD / EFFECTIVE_INST_SPEC |
| L1I cache miss rate due to prefetch access | L1I_CACHE_REFILL_PRF / EFFECTIVE_INST_SPEC |
| L1I cache miss rate due to prefetch access which was generated by software prefetch | (L1I_CACHE_REFILL_PRF - L1I_CACHE_REFILL_HWPRF) / EFFECTIVE_INST_SPEC |
| L1I cache miss rate due to prefetch access which was generated by hardware prefetch | L1I_CACHE_REFILL_HWPRF / EFFECTIVE_INST_SPEC |
| L1I cache hit rate | L1I_CACHE_HIT / EFFECTIVE_INST_SPEC |

| Indicator | Formula |
|---|---|
| L1I cache hit rate due to read access | L1I_CACHE_HIT_RD / EFFECTIVE_INST_SPEC |
| L1I cache hit (MIB match) rate due to read access | L1I_LFB_HIT_RD / EFFECTIVE_INST_SPEC |
| L1D cache miss rate | L1D_CACHE_REFILL / EFFECTIVE_INST_SPEC |
| Access rate from L1D cache due to demand access changing L1D cache status (shared → exclusive) | L1D CACHE BTC / EFFECTIVE INST SPEC |
| L1D cache miss rate due to demand access | L1D_CACHE_REFILL_DM / EFFECTIVE_INST_SPEC |
| L1D cache miss rate due to demand read access | L1D_CACHE_REFILL_DM_RD / EFFECTIVE_INST_SPEC |
| L1D cache miss rate due to demand write access | L1D_CACHE_REFILL_DM_WR / EFFECTIVE_INST_SPEC |
| L1D cache miss rate due to read access | L1D_CACHE_REFILL_RD / EFFECTIVE_INST_SPEC |
| L1D cache miss rate due to write access | L1D_CACHE_REFILL_WR / EFFECTIVE_INST_SPEC |
| L1D cache miss rate due to prefetch access | L1D_CACHE_REFILL_PRF / EFFECTIVE_INST_SPEC |
| L1D cache miss rate due to software prefetch access | (L1D_CACHE_REFILL_PRF - L1D_CACHE_REFILL_HWPRF) /EFFECTIVE_INST_SPEC |
| L1D cache miss rate due to hardware prefetch access | L1D_CACHE_REFILL_HWPRF / EFFECTIVE_INST_SPEC |
| L1D cache hit rate | L1D_CACHE_HIT / EFFECTIVE_INST_SPEC |
| L1D cache hit rate due to demand read access | L1D_CACHE_HIT_RD / EFFECTIVE_INST_SPEC |
| L1D cache hit rate due to demand write access | L1D_CACHE_HIT_WR / EFFECTIVE_INST_SPEC |
| L1D cache hit (MIB match) rate due to demand read access | L1D_LFB_HIT_RD / EFFECTIVE_INST_SPEC |
| L1D cache hit (MIB match) rate due to demand write access | L1D_LFB_HIT_WR / EFFECTIVE_INST_SPEC |
| L2 cache miss rate | L2D_CACHE_REFILL / EFFECTIVE_INST_SPEC |
| L2 cache access rate due to demand access changing L1D and L2 cache status (shared → exclusive) | L2D_CACHE_BTC / EFFECTIVE_INST_SPEC |
| L2 cache miss rate due to demand flow | L2D_CACHE_REFILL_DM / EFFECTIVE_INST_SPEC |
| L2 cache miss rate due to demand read flow | L2D_CACHE_REFILL_DM_RD / EFFECTIVE_INST_SPEC |
| L2 cache miss rate due to demand write flow | L2D_CACHE_REFILL_DM_WR / EFFECTIVE_INST_SPEC |
| L2 cache miss rate due to demand write (exclusive monitor) flow | L2D_CACHE_REFILL_DM_WR_EXCL / EFFECTIVE_INST_SPEC |
| L2 cache miss rate due to demand write (atomic operation) flow | L2D_CACHE_REFILL_DM_WR_ATOM / EFFECTIVE_INST_SPEC |
| L2 cache miss rate due to read flow | L2D_CACHE_REFILL_RD / EFFECTIVE_INST_SPEC |
| L2 cache miss rate due to write flow | L2D_CACHE_REFILL_WR / EFFECTIVE_INST_SPEC |
| L2 cache miss rate due to prefetch flow | L2D_CACHE_REFILL_PRF / EFFECTIVE_INST_SPEC |
| L2 cache miss rate due to prefetch flow which was generated by software prefetch | (L2D_CACHE_REFILL_PRF - L2D_CACHE_REFILL_HWPRF) / EFFECTIVE_INST_SPEC |
| L2 cache miss rate due to prefetch flow which was generated by hardware prefetch | L2D_CACHE_REFILL_HWPRF / EFFECTIVE_INST_SPEC |
| L2 cache hit rate | L2D CACHE HIT / EFFECTIVE INST SPEC |
| L2 cache hit rate due to demand read flow | L2D_CACHE_HIT_RD / EFFECTIVE_INST_SPEC |
| L2 cache hit rate due to demand write flow | L2D_CACHE_HIT_WR / EFFECTIVE_INST_SPEC |
| L2 cache hit (MIB match) rate due to demand read flow | L2D_LFB_HIT_RD / EFFECTIVE_INST_SPEC |
| L2 cache hit (MIB match) rate due to demand write flow | L2D_LFB_HIT_WR / EFFECTIVE_INST_SPEC |
| L3 cache miss rate | L2D_CACHE_REFILL_L3D_MISS / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to demand flow | L2D_CACHE_REFILL_L3D_MISS_DM / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to demand read flow | L2D_CACHE_REFILL_L3D_MISS_DM_RD / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to demand write flow | L2D_CACHE_REFILL_L3D_MISS_DM_WR / EFFECTIVE_INST_SPEC |

| Indicator | Formula |
|---|---|
| L3 cache miss rate due to prefetch flow | L2D_CACHE_REFILL_L3D_MISS_PRF / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to prefetch flow which was generated by software prefetch | (L2D_CACHE_REFILL_L3D_MISS_PRF - L2D_CACHE_REFILL_L3D_MISS_HWPRF) / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to prefetch flow which was generated by hardware prefetch | L2D_CACHE_REFILL_L3D_MISS_HWPRF / EFFECTIVE_INST_SPEC |
| L3 cache hit rate | L2D_CACHE_REFILL_L3D_HIT / EFFECTIVE_INST_SPEC |
| L3 cache hit rate due to demand flow | L2D_CACHE_REFILL_L3D_HIT_DM / EFFECTIVE_INST_SPEC |
| L3 cache hit rate due to demand read flow | L2D_CACHE_REFILL_L3D_HIT_DM_RD / EFFECTIVE_INST_SPEC |
| L3 cache hit rate due to demand write flow | L2D_CACHE_REFILL_L3D_HIT_DM_WR / EFFECTIVE_INST_SPEC |
| L3 cache hit rate due to prefetch flow | L2D_CACHE_REFILL_L3D_HIT_PRF / EFFECTIVE_INST_SPEC |
| L3 cache hit rate due to prefetch flow which was generated by software prefetch | (L2D_CACHE_REFILL_L3D_HIT_PRF - L2D_CACHE_REFILL_L3D_HIT_HWPRF) / EFFECTIVE_INST_SPEC |
| L3 cache hit rate due to prefetch flow which was generated by hardware prefetch | L2D_CACHE_REFILL_L3D_HIT_HWPRF / EFFECTIVE_INST_SPEC |
| L3 cache miss and prefetch buffer hit rate due to demand flow | L2D_CACHE_REFILL_L3D_MISS_DM_PFTGT_HIT / EFFECTIVE_INST_SPEC |
| L3 cache miss and prefetch buffer hit rate due to demand read flow | L2D_CACHE_REFILL_L3D_MISS_DM_RD_PFTGT_HIT / EFFECTIVE_INST_SPEC |
| L3 cache miss and prefetch buffer hit rate due to demand write flow | L2D_CACHE_REFILL_L3D_MISS_DM_WR_PFTGT_HIT / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to demand flow accessing memory in the same socket | L2D_CACHE_REFILL_L3D_MISS_DM_L_MEM / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to demand flow accessing memory in different socket | L2D_CACHE_REFILL_L3D_MISS_DM_FR_MEM / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to demand flow where the requests access the different L2 cache from the request in the same Cluster as the requests in the same COD/SRD set as the requests | L2D_CACHE_REFILL_L3D_MISS_DM_L_L2 / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to demand flow where the requests access the L2 cache in the different COD/SRD set from the request in the same socket as the requests or in the different Cluster from the request in the same COD/SRD set as the requests | L2D_CACHE_REFILL_L3D_MISS_DM_NR_L2 / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to demand flow where the requests access the L3 cache in the different COD/SRD set from the request in the same socket as the requests or in the different Cluster from the request in the same COD/SRD set as the requests | L2D_CACHE_REFILL_L3D_MISS_DM_NR_L3 / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to demand flow accessing L2 cache in different sockets | L2D_CACHE_REFILL_L3D_MISS_DM_FR_L2 / EFFECTIVE_INST_SPEC |
| L3 cache miss rate due to demand flow accessing L3 cache in different sockets | L2D_CACHE_REFILL_L3D_MISS_DM_FR_L3 / EFFECTIVE_INST_SPEC |
| Average latency of L1I cache miss processing | L1I_CACHE_REFILL_PERCYC / L1I_CACHE_REFILL |
| Average latency of L1D cache miss processing | L1D_CACHE_REFILL_PERCYC / L1D_CACHE_REFILL |
| Average latency of L2 cache miss processing | L2D_CACHE_REFILL_PERCYC / L2D_CACHE_REFILL |
| Average number of outstanding misses in L1I cache miss processing | L1I_CACHE_REFILL_PERCYC / CPU_CYCLES |
| Average number of outstanding misses in L1D cache miss processing | L1D_CACHE_REFILL_PERCYC / CPU_CYCLES |

| Indicator | Formula |
|---|---|
| Average number of outstanding misses in L2D cache miss processing | L2D_CACHE_REFILL_PERCYC / CPU_CYCLES |
| L1-ITLB miss rate | L1I_TLB_REFILL / EFFECTIVE_INST_SPEC |
| L1-DTLB miss rate | L1D_TLB_REFILL / EFFECTIVE_INST_SPEC |
| L2-ITLB miss rate | L2I_TLB_REFILL / EFFECTIVE_INST_SPEC |
| L2-DTLB miss rate | L2D_TLB_REFILL / EFFECTIVE_INST_SPEC |
| Bidirectional effective bandwidth between L1D cache and L2 cache | (L1D_CACHE_REFILL + L1D_CACHE_WB) * Line Size * Processor frequency / CPU_CYCLES |
| Bidirectional effective bandwidth between L2 cache and L3 cache | (L2D_CACHE_REFILL + L2D_CACHE_WB) * Line Size * Processor frequency / CPU_CYCLES |
| Address arithmetic pipeline A busy rate | EAGA_VAL / CPU_CYCLES |
| Address arithmetic pipeline B busy rate | EAGB_VAL / CPU_CYCLES |
| Predicate arithmetic pipeline busy rate | PRX_VAL / CPU_CYCLES |
| Fixed-point arithmetic pipeline A busy rate | EXA_VAL / CPU_CYCLES |
| Fixed-point arithmetic pipeline B busy rate | EXB_VAL / CPU_CYCLES |
| Fixed-point arithmetic pipeline C busy rate | EXC_VAL / CPU_CYCLES |
| Fixed-point arithmetic pipeline D busy rate | EXD_VAL / CPU_CYCLES |
| Floating-point arithmetic pipeline A busy rate | FLA_VAL / CPU_CYCLES |
| Floating-point arithmetic pipeline B busy rate | FLB_VAL / CPU_CYCLES |
| Floating-point store pipeline A busy rate | STEA_VAL / CPU_CYCLES |
| Floating-point store pipeline B busy rate | STEB_VAL / CPU_CYCLES |
| Floating-point store pipeline busy rate | STFL_VAL / CPU_CYCLES |
| Predicate store pipeline busy rate | STPX_VAL / CPU_CYCLES |
| gather 2flow rate | L1_PIPE_COMP_GATHER_2FLOW / (L1_PIPE0_COMP + L1_PIPE1_COMP) |
| gather 1flow rate | L1_PIPE_COMP_GATHER_1FLOW / (L1_PIPE0_COMP + L1_PIPE1_COMP) |
| gather 0flow rate | L1_PIPE_COMP_GATHER_0FLOW / (L1_PIPE0_COMP + L1_PIPE1_COMP) |
| L1 PIPE0 busy rate | L1_PIPE0_VAL / CPU_CYCLES |
| L1 PIPE1 busy rate | L1_PIPE1_VAL / CPU_CYCLES |
| L1 PIPE2 busy rate | L1_PIPE2_VAL / CPU_CYCLES |
| FLA pipeline predicate rate | FLA_VAL_PRD_CNT / FLA_VAL_FOR_PRD / 32 |
| FLB pipeline predicate rate | FLB_VAL_PRD_CNT / FLB_VAL_FOR_PRD / 32 |
| L1 PIPE0 SEC0 rate | L1_PIPE0_VAL_IU_NOT_SEC0 / L1_PIPE0_VAL |
| L1 PIPE0 SEC1 rate | (L1_PIPE0_VAL - L1_PIPE0_VAL_IU_NOT_SEC0) / L1_PIPE0_VAL |
| L1 PIPE1 SEC0 rate | L1_PIPE1_VAL_IU_NOT_SEC0 / L1_PIPE1_VAL |
| L1 PIPE1 SEC1 rate | (L1_PIPE1_VAL - L1_PIPE1_VAL_IU_NOT_SEC0) / L1_PIPE1_VAL |
| Speculative execution rate | (OP_SPEC - OP_RETIRED) / OP_RETIRED |

# 1.4. Cycle Accounting

One of the processor performance indicators is cycles per instruction (CPI), which represents the average CPU cycles spent by the processor to execute one instruction. Here, CPI can be considered as the accumulated processing time of various operation-flows for instruction execution; for example, time for operations and memory access. The expression of CPI as an accumulation of such individual processing times is called "Cycle Accounting." The MONAKA has events for Cycle Accounting. Table 1-6 summarizes the events. Like events for Instruction Mix, there is an inclusion relationship between objects measured. "Other" indicates an item that can be calculated by combining events. The formulas are shown in Table 1-7.

**Table 1-6 Performance Events for Cycle Accounting**

| Events for Cycle Accounting | Target of Counting by Event |
|---|---|
| CPU_CYCLES | CPU clock cycles |
| _0INST_COMMIT | Cycles during which number of instruction commits is 0 |
| LD_COMP_WAIT | Cycles during which oldest instruction in CSE cannot be committed due to wait for completion of memory access |
| LD_COMP_WAIT_L2_MISS | Cycles during L2 cache miss, taken out of LD_COMP_WAIT |
| LD_COMP_WAIT_L2_MISS_EX | Cycles caused by instructions belonging to Base Instructions, taken out of LD_COMP_WAIT_L2_MISS |
| Other (ld_comp_wait_l2_miss_fl) | Cycles caused by instructions belonging to SIMD&FP or SVE instruction, taken out of LD_COMP_WAIT_L2_MISS |
| LD_COMP_WAIT_L1_MISS | Cycles during L1D cache miss and L2 cache hit, taken out of LD_CIMP_WAIT (Strictly speaking, this includes the cycles until the L2 cache miss is determined at the L2 cache miss time.) |
| LD_COMP_WAIT_L1_MISS_EX | Cycles caused by instructions belonging to Base Instructions, taken out of LD_COMP_WAIT_L1_MISS |
| Other (ld_comp_wait_l1_miss_fl) | Cycles caused by instructions belonging to SIMD&FP or SVE instruction, taken out of LD_COMP_WAIT_L1_MISS |
| LD_COMP_WAIT_PFP_BUSY | Cycles during which memory access instructions cannot be committed due to insufficient resources for L2 cache prefetch processing, taken out of LD_COMP_WAIT (This represents an event where prefetch flow cannot be processed and the flow generator instruction cannot be committed) |
| LD_COMP_WAIT_PFP_BUSY_EX | Cycles caused by instructions belonging to Base Instructions, taken out of LD_COMP_WAIT_PFP_BUSY |
| LD_COMP_WAIT_PFP_BUSY_SWPF | Cycles caused by software prefetch instruction, taken out of LD_COMP_WAIT_PFP_BUS |
| Other (ld_comp_wait_pfp_busy_fl) | Cycles caused by SIMD&FP or SVE instruction, taken out of LD_COMP_WAIT_PFP_BUS |
| Other (ld_comp_wait_l1_hit) | Cycles during L1D cache hit, taken out of LD_COMP_WAIT (Strictly speaking, this includes the cycles until the L1D cache miss is determined at the L1D cache miss time.) |
| Other (ld_comp_wait_l1_hit_ex) | Cycles caused by instructions belonging to Base Instructions, taken out of ld_comp_wait_l1_hit |
| Other (ld_comp_wait_l1_hit_fl) | Cycles caused by instructions that belong to SIMD&FP or SVE instructions, taken out of ld_comp_wait_l1_hit |
| EU_COMP_WAIT | Cycles during which oldest instruction in CSE cannot be committed due to wait for completion of operations |
| FL_COMP_WAIT | Cycles caused by instructions belonging to SIMD&FP or SVE instruction, taken out of EU_COMP_WAIT |

| Events for Cycle Accounting | Target of Counting by Event |
|---|---|
| Other (ex_comp_wait) | Cycles caused by instructions belonging to Base Instructions, taken out of EU_COMP_WAIT |
| BR_COMP_WAIT | Cycles during which oldest instruction in CSE cannot be committed because it is branch direction and waiting for branch direction determination |
| ROB_EMPTY | Cycles during which instructions cannot be committed because CSE is empty (The instruction does not exist after the decode stage) |
| ROB_EMPTY_STQ_BUSY | Cycles during which instructions cannot be committed because CSE is empty and Virtual SP is full (Since Virtual SP is full, decoding is currently stopped. |
| WFE_WFI_CYCLE | Cycles during which behavior of PE is stopped by WFE instruction or WFI (This appears as a synchronization wait time in a parallel program) |
| RETENTION_CYCLE | Cycle during which behavior of PE is stopped by RETENTION state (This appears as time in low power states) |
| Other (rob_empty_not_stq_busy) | Cycles during which instructions cannot be committed because CSE is empty due to other causes (This mainly appears as the instruction fetch time) |
| UOP_ONLY_COMMIT | Cycles during which only μOP instructions are committed (For an architecture instruction that is decoded into 2 or more μOP instructions, commit of the last μOP instruction means commit of the architecture instruction. This means there is a condition where only μOP instructions are committed) |
| SINGLE_MOVPRFX_COMMIT | Cycles during which only unpacked MOVPRFX instructions are committed |
| Other (0inst_commit_other) | Cycles during which instructions cannot be committed due to other causes |
| _1INST_COMMIT | Cycles during which number of instruction commits is 1 |
| _2INST_COMMIT | Cycles during which number of instruction commits is 2 |
| _3INST_COMMIT | Cycles during which number of instruction commits is 3 |
| _4INST_COMMIT | Cycles during which number of instruction commits is 4 |
| _5INST_COMMIT | Cycles during which number of instruction commits is 5 |
| other (6_10inst_commit) | Cycles during which number of instruction commits is 6 or more |
| **Event Independent of Inclusion Relationship** | |
| LD_COMP_WAIT_EX | Cycles caused by instructions belonging to Base Instructions, taken out of LD_COMP_WAIT |

**Table 1-7 Formulas for Other (Cycle Accounting)**

| Item | Formula |
|---|---|
| ld_comp_wait_l2_miss_fl | LD_COMP_WAIT_L2_MISS − LD_COMP_WAIT_L2_MISS_EX |
| ld_comp_wait_l1_miss_fl | LD_COMP_WAIT_L1_MISS − LD_COMP_WAIT_L1_MISS_EX |
| ld_comp_wait_pfp_busy_fl | LD_COMP_WAIT_PFP_BUSY<br>– (LD_COMP_WAIT_PFP_BUSY_EX + LD_COMP_WAIT_PFP_BUSY_SWPF) |
| ld_comp_wait_l1_hit | LD_COMP_WAIT<br>– (LD_COMP_WAIT_L2_MISS + LD_COMP_WAIT_L1_MISS<br>+ LD_COMP_WAIT_PFP_BUSY) |
| ld_comp_wait_l1_hit_ex | LD_COMP_WAIT_EX<br>−(LD_COMP_WAIT_L2_MISS_EX + LD_COMP_WAIT_L1_MISS_EX<br>+ LD_COMP_WAIT_PFP_BUSY_EX) |
| ld_comp_wait_l1_hit_fl | ld_comp_wait_l1_hit - ld_comp_wait_l1_hit_ex |
| ex_comp_wait | EU_COMP_WAIT - FL_COMP_WAIT |
| rob_empty_not_stq_busy | ROB_EMPTY<br>– (ROB_EMPTY_STQ_BUSY + WFE_WFI_CYCLE + RETENTION_CYCLE) |

| Item | Formula |
| --- | --- |
| 0inst_commit_other | _0INST_COMMIT<br>– (UOP_ONLY_COMMIT + SINGLE_MOVPRFX_COMMIT<br>+ LD_COMP_WAIT + EU_COMP_WAIT + BR_COMP_WAIT + ROB_EMPTY) |
| 6_10inst_commit | CPU_CYCLES<br>– (_0INST_COMMIT + _1INST_COMMIT + _2INST_COMMIT<br>+ _3INST_COMMIT + _4INST_COMMIT + _5INST_COMMIT) |

# 1.5. Decode base Cycle Accounting

MONAKA also implements Decode-based Cycle Accounting as a metric similar to the Cycle Accounting described in 1.4. The difference between Cycle Accounting and Decode-based Cycle Accounting is that the former measures the breakdown of cycles at the commit stage, whereas the latter measures the breakdown of cycles at the decode stage. While Cycle Accounting can measure stall conditions that are more directly tied to performance, Decode-based Cycle Accounting can measure hidden stall conditions that are not directly visible in performance. Users can choose between them as needed.

Table Table 1-8 summarizes the events for Decode-based Cycle Accounting implemented in MONAKA. "Other" represents items that can be derived by combining events, and the formulas for these are summarized in Table 1-9.

**Table 1-8 Performance Events for Decode base Cycle Accounting**

| Events for Decode base Cycle Accounting | Target of Counting by Event |
|---|---|
| CPU_CYCLES | CPU clock cycles |
| STALL | Every cycle that no instruction was dispatched from decode unit |
| STALL_FRONTEND | Every cycle counted by the CPU_CYCLES event on that no operation was issued because there are no operations available to issue for this PE from the frontend |
| STALL_FRONTEND_MEMBOUND | Every cycle counted by STALL_FRONTEND when no instructions are delivered from the memory system |
| STALL_FRONTEND_L1I | Every cycle counted by STALL_FRONTEND_MEMBOUND when there is a demand instruction miss in the first level of instruction cache |
| STALL_FRONTEND_L2I | Every cycle counted by STALL_FRONTEND_MEMBOUND when there is a demand instruction miss in the second level of instruction cache |
| STALL_FRONTEND_MEM | Every cycle counted by STALL_FRONTEND_MEMBOUND when there is a demand instruction miss in the last level of instruction cache within the PE clock domain or a non-cacheable instruction fetch in progress |
| STALL_FRONTEND_TLB | Every cycle counted by STALL_FRONTEND_MEMBOUND when there is a demand instruction miss in the instruction TLB |
| STALL_FRONTEND_CPUBOUND | Every cycle counted by STALL_FRONTEND when the frontend is stalled on a frontend processor resource, not including memory |
| STALL_FRONTEND_FLUSH | Every cycle counted by STALL_FRONTEND_CPUBOUND when the frontend is recovering from a pipeline flush |
| Other (stall_frontend_cpubound_other) | Every cycle counted by "stall_frontend_cpubound_other" , described in Table 1-9 |
| STALL_BACKEND | Every cycle counted by the CPU_CYCLES event on that no operation was issued because the backend is unable to accept any operation. |
| STALL_BACKEND_MEMBOUND | Every cycle counted by STALL_BACKEND when the backend is waiting for a memory access to complete. |
| STALL_BACKEND_L1D | Every cycle counted by STALL_BACKEND_MEMBOUND when there is a demand data miss in L1D cache |
| STALL_BACKEND_L2D | Every cycle counted by STALL_BACKEND_MEMBOUND when there is a demand data miss in L2 cache |
| STALL_BACKEND_MEM | Every cycle that no instruction was dispatched from decode unit due to memory stall |
| STALL_BACKEND_TLB | Every cycle counted by STALL_BACKEND_MEMBOUND when there is a demand data miss in the data TLB |

| Events for Decode base Cycle Accounting | Target of Counting by Event |
|---|---|
| STALL_BACKEND_CPUBOUND | Every cycle counted by STALL_BACKEND when the backend is stalled on a processor resource, not including memory |
| STALL_BACKEND_BUSY | Every cycle counted by STALL_BACKEND when operations are available from the frontend but the backend is not able to accept an operation because an execution unit is busy. |
| STALL_BACKEND_RENAME | Every cycle counted by STALL_BACKEND_CPUBOUND when operations are available from the frontend but at least one is not ready to be sent to the backend because no rename register is available. |
| Other (stall_backend_busy_other) | Every cycle counted by “stall_backend_busy_other” , described in Table 1-9 |
| Other (stall_backend_cpubound_other) | Every cycle counted by “stall_backend_cpubound other” , described in Table 1-9 |
| Other (stall_backend_other) | Every cycle counted by “stall_backend_other” , described in Table 1-9 |
| Other (dispatched_cycles) | Every cycle counted by “dispatched cycles” , described in Table 1-9 |

**Table 1-9 Formulas for Other (Decode base Cycle Accounting)**

| Item | Formula |
|---|---|
| Other (stall_frontend_cpubound_other) | STALL_FRONTEND_CPUBOUND - STALL_FRONTEND_FLUSH |
| Other (stall_backend_busy_other) | STALL_BACKEND_BUSY - STALL_BACKEND_RENAME |
| Other (stall_backend_cpubound_other) | STALL_BACKEND_CPUBOUND - STALL_BACKEND_BUSY |
| Other (stall_backend_other) | STALL_BACKEND - (STALL_BACKEND_MEMBOUND + STALL_BACKEND_CPUBOUND) |
| Other (dispatched_cycles) | CPU_CYCLES - STALL |

# 1.6. Hardware Prefetch

**Table 1-10 Performance Events for Hardware Prefetch**

| Performance Event | Description of Event |
|---|---|
| L1HWPF_STREAM_PF | Number of streaming prefetch requests to L1D cache generated by hardware prefetcher |
| L1HWPF_STRIDE_PF | Number of stride prefetch requests to L1D cache generated by hardware prefetcher |
| L1HWPF_PFTGT_PF | Number of LDS prefetch requests to L1D cache generated by hardware prefetcher |
| L2HWPF_STREAM_PF | Number of streaming prefetch requests to L2 cache generated by hardware prefetcher |
| L2HWPF_STRIDE_PF | Number of stride prefetch requests to L2 cache generated by hardware prefetcher |
| L2HWPF_OTHER | Number of prefetch requests to L2 cache generated by the other causes |
| L3HWPF_STREAM_PF | Number of streaming prefetch requests to L3 cache generated by hardware prefetcher |
| L3HWPF_STRIDE_PF | Number of stride prefetch requests to L3 cache generated by hardware prefetcher |
| L3HWPF_OTHER | Number of prefetch requests to L3 cache generated by the other causes |

**Table 1-11 Formulas for Hardware Prefetch**

| Item | Formula |
|---|---|
| L1 HWPF STREAM PF rate | L1HWPF_STREAM_PF / (L1HWPF_STREAM_PF + L1HWPF_STRIDE_PF + L1HWPF_PFTGT_PF) |
| L1 HWPF STRIDE PF rate | L1HWPF_STRIDE_PF / (L1HWPF_STREAM_PF + L1HWPF_STRIDE_PF + L1HWPF_PFTGT_PF) |
| L1 HWPF PFTGT PF rate | L1HWPF_PFTGT_PF / (L1HWPF_STREAM_PF + L1HWPF_STRIDE_PF + L1HWPF_PFTGT_PF) |
| L2 HWPF STREAM PF rate | L2HWPF_STREAM_PF / (L2HWPF_STREAM_PF + L2HWPF_STRIDE_PF + L2HWPF_OTHER) |
| L2 HWPF STRIDE PF rate | L2HWPF_STRIDE_PF / (L2HWPF_STREAM_PF + L2HWPF_STRIDE_PF + L2HWPF_OTHER) |
| L2 HWPF OTHER rate | L2HWPF_OTHER / (L2HWPF_STREAM_PF + L2HWPF_STRIDE_PF + L2HWPF_OTHER) |
| L3 HWPF STREAM PF rate | L3HWPF_STREAM_PF / (L3HWPF_STREAM_PF + L3HWPF_STRIDE_PF + L3HWPF_OTHER) |
| L3 HWPF STRIDE PF rate | L3HWPF_STRIDE_PF / (L3HWPF_STREAM_PF + L3HWPF_STRIDE_PF + L3HWPF_OTHER) |
| L3 HWPF OTHER rate | L3HWPF_OTHER / (L3HWPF_STREAM_PF + L3HWPF_STRIDE_PF + L3HWPF_OTHER) |

# 1.7. Predicate Rate

**Table 1-12 Performance Events for Predicate Rate**

| Performance Event | Description of Event |
|---|---|
| FLA_VAL_PRD_CNT | Number of 1's in the predicate bits of request in FLA pipeline, where it is corrected so that it becomes 32 when all bits are 1 |
| FLB_VAL_PRD_CNT | Number of 1's in the predicate bits of request in FLB pipeline, where it is corrected so that it becomes 32 when all bits are 1 |
| FLA_VAL_FOR_PRD | Number of valid cycles of FLA pipeline |
| FLB_VAL_FOR_PRD | Number of valid cycles of FLA pipeline |
| L1_PIPE0_COMP_PRD_CNT | Number of 1's in the predicate bits of request in L1D cache pipeline#0, where it is corrected so that it becomes 32 when all bits are 1 |
| L1_PIPE1_COMP_PRD_CNT | Number of 1's in the predicate bits of request in L1D cache pipeline#1, where it is corrected so that it becomes 32 when all bits are 1 |
| L1_PIPE0_COMP | Number of completed requests in L1D cache pipeline#0 |
| L1_PIPE1_COMP | Number of completed requests in L1D cache pipeline#1 |

**Table 1-13 Formulas for Predicate Rate**

| Item | Formula |
|---|---|
| FLA pipeline predicate rate | FLA_VAL_PRD_CNT / FLA_VAL_FOR_PRD / 32 |
| FLB pipeline predicate rate | FLB_VAL_PRD_CNT / FLB_VAL_FOR_PRD / 32 |
| L1 PIPE0 predicate rate | L1_PIPE0_COMP_PRD_CNT / L1_PIPE0_COMP / 32 |
| L1 PIPE1 predicate rate | L1_PIPE1_COMP_PRD_CNT / L1_PIPE1_COMP / 32 |

# 1.8. Breakdown of TLB Page Size

**Table 1-14 Performance Events for Breakdown of TLB Page Size**

| Performance Event | Description of Event |
|---|---|
| L1I_TLB_4K | Number of operations that cause a TLB access to the L1I in 4KB page |
| L1I_TLB_64K | Number of operations that cause a TLB access to the L1I in 64KB page |
| L1I_TLB_2M | Number of operations that cause a TLB access to the L1I in 2MB page |
| L1I_TLB_32M | Number of operations that cause a TLB access to the L1I in 32MB page |
| L1I_TLB_512M | Number of operations that cause a TLB access to the L1I in 512MB page |
| L1I_TLB_1G | Number of operations that cause a TLB access to the L1I in 1GB page |
| L1I_TLB_16G | Number of operations that cause a TLB access to the L1I in 16GB page |
| L1D_TLB_4K | Number of operations that cause a TLB access to the L1D in 4KB page |
| L1D_TLB_64K | Number of operations that cause a TLB access to the L1D in 64KB page |
| L1D_TLB_2M | Number of operations that cause a TLB access to the L1D in 2MB page |
| L1D_TLB_32M | Number of operations that cause a TLB access to the L1D in 32MB page |
| L1D_TLB_512M | Number of operations that cause a TLB access to the L1D in 512MB page |
| L1D_TLB_1G | Number of operations that cause a TLB access to the L1D in 1GB page |
| L1D_TLB_16G | Number of operations that cause a TLB access to the L1D in 16GB page |
| L1I_TLB_REFILL_4K | Number of operations that cause a TLB refill of the L1I in 4KB page |
| L1I_TLB_REFILL_64K | Number of operations that cause a TLB refill of the L1I in 64KB page |
| L1I_TLB_REFILL_2M | Number of operations that cause a TLB refill of the L1I in 2MB page |
| L1I_TLB_REFILL_32M | Number of operations that cause a TLB refill of the L1I in 32MB page |
| L1I_TLB_REFILL_512M | Number of operations that cause a TLB refill of the L1I in 512MB page |
| L1I_TLB_REFILL_1G | Number of operations that cause a TLB refill of the L1I in 1GB page |
| L1I_TLB_REFILL_16G | Number of operations that cause a TLB refill of the L1I in 16GB page |
| L1D_TLB_REFILL_4K | Number of operations that cause a TLB refill of the L1D in 4KB page |
| L1D_TLB_REFILL_64K | Number of operations that cause a TLB refill of the L1D in 64KB page |
| L1D_TLB_REFILL_2M | Number of operations that cause a TLB refill of the L1D in 2MB page |
| L1D_TLB_REFILL_32M | Number of operations that cause a TLB refill of the L1D in 32MB page |
| L1D_TLB_REFILL_512M | Number of operations that cause a TLB refill of the L1D in 512MB page |
| L1D_TLB_REFILL_1G | Number of operations that cause a TLB refill of the L1D in 1GB page |
| L1D_TLB_REFILL_16G | Number of operations that cause a TLB refill of the L1D in 16GB page |
| L2I_TLB_4K | Number of operations that cause a TLB access to the L2I in 4KB page |
| L2I_TLB_64K | Number of operations that cause a TLB access to the L2I in 64KB page |
| L2I_TLB_2M | Number of operations that cause a TLB access to the L2I in 2MB page |
| L2I_TLB_32M | Number of operations that cause a TLB access to the L2I in 32MB page |
| L2I_TLB_512M | Number of operations that cause a TLB access to the L2I in 512MB page |
| L2I_TLB_1G | Number of operations that cause a TLB access to the L2I in 1GB page |
| L2I_TLB_16G | Number of operations that cause a TLB access to the L2I in 16GB page |
| L2D_TLB_4K | Number of operations that cause a TLB access to the L2D in 4KB page |
| L2D_TLB_64K | Number of operations that cause a TLB access to the L2D in 64KB page |
| L2D_TLB_2M | Number of operations that cause a TLB access to the L2D in 2MB page |
| L2D_TLB_32M | Number of operations that cause a TLB access to the L2D in 32MB page |
| L2D_TLB_512M | Number of operations that cause a TLB access to the L2D in 512MB page |
| L2D_TLB_1G | Number of operations that cause a TLB access to the L2D in 1GB page |
| L2D_TLB_16G | Number of operations that cause a TLB access to the L2D in 16GB page |
| L2I_TLB_REFILL_4K | Number of operations that cause a TLB refill of the L2I in 4KB page |

| Performance Event | Description of Event |
|---|---|
| L2I_TLB_REFILL_64K | Number of operations that cause a TLB refill of the L2I in 64KB page. |
| L2I_TLB_REFILL_2M | Number of operations that cause a TLB refill of the L2I in 2MB page. |
| L2I_TLB_REFILL_32M | Number of operations that cause a TLB refill of the L2I in 32MB page. |
| L2I_TLB_REFILL_512M | Number of operations that cause a TLB refill of the L2I in 512MB page. |
| L2I_TLB_REFILL_1G | Number of operations that cause a TLB refill of the L2I in 1GB page. |
| L2I_TLB_REFILL_16G | Number of operations that cause a TLB refill of the L2I in 16GB page. |
| L2D_TLB_REFILL_4K | Number of operations that cause a TLB refill of the L2D in 4KB page. |
| L2D_TLB_REFILL_64K | Number of operations that cause a TLB refill of the L2D in 64KB page. |
| L2D_TLB_REFILL_2M | Number of operations that cause a TLB refill of the L2D in 2MB page. |
| L2D_TLB_REFILL_32M | Number of operations that cause a TLB refill of the L2D in 32MB page. |
| L2D_TLB_REFILL_512M | Number of operations that cause a TLB refill of the L2D in 512MB page. |
| L2D_TLB_REFILL_1G | Number of operations that cause a TLB refill of the L2D in 1GB page. |
| L2D_TLB_REFILL_16G | Number of operations that cause a TLB refill of the L2D in 16GB page. |

**Table 1-15 Formulas for Breakdown of TLB Page size**

| Item | Formula |
|---|---|
| L1I TLB 4K page rate | L1I_TLB_4K / L1I_TLB |
| L1I TLB 64K page rate | L1I_TLB_64K / L1I_TLB |
| L1I TLB 2M page rate | L1I_TLB_2M / L1I_TLB |
| L1I TLB 32M page rate | L1I_TLB_32M / L1I_TLB |
| L1I TLB 512M page rate | L1I_TLB_512M / L1I_TLB |
| L1I TLB 1G page rate | L1I_TLB_1G / L1I_TLB |
| L1I TLB 16G page rate | L1I_TLB_16G / L1I_TLB |
| L1D TLB 4K page rate | L1D_TLB_4K / L1D_TLB |
| L1D TLB 64K page rate | L1D_TLB_64K / L1D_TLB |
| L1D TLB 2M page rate | L1D_TLB_2M / L1D_TLB |
| L1D TLB 32M page rate | L1D_TLB_32M / L1D_TLB |
| L1D TLB 512M page rate | L1D_TLB_512M / L1D_TLB |
| L1D TLB 1G page rate | L1D_TLB_1G / L1D_TLB |
| L1D TLB 16G page rate | L1D_TLB_16G / L1D_TLB |
| L1I TLB REFILL 4K page rate | L1I_TLB_REFILL_4K / L1I_TLB_REFILL |
| L1I TLB REFILL 64K page rate | L1I_TLB_REFILL_64K / L1I_TLB_REFILL |
| L1I TLB REFILL 2M page rate | L1I_TLB_REFILL_2M / L1I_TLB_REFILL |
| L1I TLB REFILL 32M page rate | L1I_TLB_REFILL_32M / L1I_TLB_REFILL |
| L1I TLB REFILL 512M page rate | L1I_TLB_REFILL_512M / L1I_TLB_REFILL |
| L1I TLB REFILL 1G page rate | L1I_TLB_REFILL_1G / L1I_TLB_REFILL |
| L1I TLB REFILL 16G page rate | L1I_TLB_REFILL_16G / L1I_TLB_REFILL |
| L1D TLB REFILL 4K page rate | L1D_TLB_REFILL_4K / L1D_TLB_REFILL |
| L1D TLB REFILL 64K page rate | L1D_TLB_REFILL_64K / L1D_TLB_REFILL |
| L1D TLB REFILL 2M page rate | L1D_TLB_REFILL_2M / L1D_TLB_REFILL |
| L1D TLB REFILL 32M page rate | L1D_TLB_REFILL_32M / L1D_TLB_REFILL |
| L1D TLB REFILL 512M page rate | L1D_TLB_REFILL_512M / L1D_TLB_REFILL |
| L1D TLB REFILL 1G page rate | L1D_TLB_REFILL_1G / L1D_TLB_REFILL |
| L1D TLB REFILL 16G page rate | L1D_TLB_REFILL_16G / L1D_TLB_REFILL |
| L2I TLB 4K page rate | L2I_TLB_4K / L2I_TLB |
| L2I TLB 64K page rate | L2I_TLB_64K / L2I_TLB |
| L2I TLB 2M page rate | L2I_TLB_2M / L2I_TLB |

| Item | Formula |
|---|---|
| L2I TLB 32M page rate | L2I_TLB_32M / L2I_TLB |
| L2I TLB 512M page rate | L2I_TLB_512M / L2I_TLB |
| L2I TLB 1G page rate | L2I_TLB_1G / L2I_TLB |
| L2I TLB 16G page rate | L2I_TLB_16G / L2I_TLB |
| L2D TLB 4K page rate | L2D_TLB_4K / L2D_TLB |
| L2D TLB 64K page rate | L2D_TLB_64K / L2D_TLB |
| L2D TLB 2M page rate | L2D_TLB_2M / L2D_TLB |
| L2D TLB 32M page rate | L2D_TLB_32M / L2D_TLB |
| L2D TLB 512M page rate | L2D_TLB_512M / L2D_TLB |
| L2D TLB 1G page rate | L2D_TLB_1G / L2D_TLB |
| L2D TLB 16G page rate | L2D_TLB_16G / L2D_TLB |
| L2I TLB REFILL 4K page rate | L2I_TLB_REFILL_4K / L2I_TLB_REFILL |
| L2I TLB REFILL 64K page rate | L2I_TLB_REFILL_64K / L2I_TLB_REFILL |
| L2I TLB REFILL 2M page rate | L2I_TLB_REFILL_2M / L2I_TLB_REFILL |
| L2I TLB REFILL 32M page rate | L2I_TLB_REFILL_32M / L2I_TLB_REFILL |
| L2I TLB REFILL 512M page rate | L2I_TLB_REFILL_512M / L2I_TLB_REFILL |
| L2I TLB REFILL 1G page rate | L2I_TLB_REFILL_1G / L2I_TLB_REFILL |
| L2I TLB REFILL 16G page rate | L2I_TLB_REFILL_16G / L2I_TLB_REFILL |
| L2D TLB REFILL 4K page rate | L2D_TLB_REFILL_4K / L2D_TLB_REFILL |
| L2D TLB REFILL 64K page rate | L2D_TLB_REFILL_64K / L2D_TLB_REFILL |
| L2D TLB REFILL 2M page rate | L2D_TLB_REFILL_2M / L2D_TLB_REFILL |
| L2D TLB REFILL 32M page rate | L2D_TLB_REFILL_32M / L2D_TLB_REFILL |
| L2D TLB REFILL 512M page rate | L2D_TLB_REFILL_512M / L2D_TLB_REFILL |
| L2D TLB REFILL 1G page rate | L2D_TLB_REFILL_1G / L2D_TLB_REFILL |
| L2D TLB REFILL 16G page rate | L2D_TLB_REFILL_16G / L2D_TLB_REFILL |

# 1.9. Exception

**Table 1-16 Performance Events for Exception**

| Events for Exception | | Target of Counting by Event |
|---|---|---|
| EXC_TAKEN | | Number of each exception taken |
| | EXC_UNDEF | Number of only other synchronous exceptions that are taken locally |
| | EXC_SVC | Number of only Supervisor Call exceptions that are taken locally |
| | EXC_PABORT | Number of only Instruction Abort exceptions that are taken locally |
| | EXC_DABORT | Number of only Data Abort or SError interrupt exceptions that are taken locally |
| | EXC_IRQ | Number of only IRQ exceptions that are taken locally, including Virtual IRQ exceptions |
| | EXC_FIQ | Number of only FIQ exceptions that are taken locally, including Virtual FIQ exceptions |
| | EXC_SMC | Number of only Secure Monitor Call exceptions. This event does not increment on SMC instructions trapped as a Hyp Trap exception |
| | EXC_HVC | Number of both Hypervisor Call exceptions taken locally in the hypervisor and those taken as an exception from Non-secure EL1 |
| | Other (exc_taken_other) | Number of exception taken counted by “exc_taken_other” , described in Table 1-17 |
| **Event Independent of Inclusion Relationship** | | |
| EXC_RETURN | | Number of each executed exception return instruction |

**Table 1-17 Formulas for Exception**

| Item | Formula |
|---|---|
| Other (exc_taken_other) | EXC_TAKEN - (EXC_UNDEF + EXC_SVC + EXC_PABORT + EXC_DABORT + EXC_IRQ + EXC_FIQ + EXC_SMC + EXC_HVC) |